

RF transmitter. An ASK based RF transmitter module with 433MHz carrier frequency is used to modulate and transmit the sensor data to the receiver.

Battery. A 6V or 9V battery is used to provide power supply to a complete sensor node.

An LED blinks to indicate that sensor node is active, and it is transmitting data.

Remote data receiver has only RF receiver module and microcontroller.

RF receiver. An ASK based RF receiver module with 433MHz carrier frequency is used to demodulate and receive data transmitted by sensor node's RF transmitter module.

Microcontroller. Arduino Nano board is used as microcontroller that gets the data from RF receiver module and gives it to computer, which stores it for future use.

LED in receiver blinks to indicate the receiver is active and it is receiving data.

Sensor node (transmitter)

As shown in circuit diagram of a sensor node in Fig. 2, there are only five major components in the transmitter circuit:

1. Sensor DHT11 that has three interfacing pins: Vcc, Gnd, and Data Out. Vcc pin is given 5V supply from Arduino board and Gnd pin is connected to common ground. Data Out pin is connected to digital pin D10 of Arduino Nano board. It is pulled high by pull-up resistor R3, as shown in Fig. 2.

2. Soil moisture sensor SS1 that has three pins for interfacing: Vcc, Gnd, and A0. Vcc pin is given 5V supply from Arduino Nano board and Gnd pin is connected to common ground. SIG/A0 pin is analogue output of sensor that is connected to analogue input pin A1 of Arduino Nano board.

3. LDR1, which is connected in pulled down configuration with a 10-kilo-ohm resistor (R2). Its output is connected to analogue input pin A0 of Arduino Nano board.

4. 433MHz RF transmitter module (TX1) that has four interfacing pins: Vcc, Gnd, Data, and Antenna. Vcc pin is connected to 5V output of Arduino board and Gnd is connected to common ground. Data pin is con-

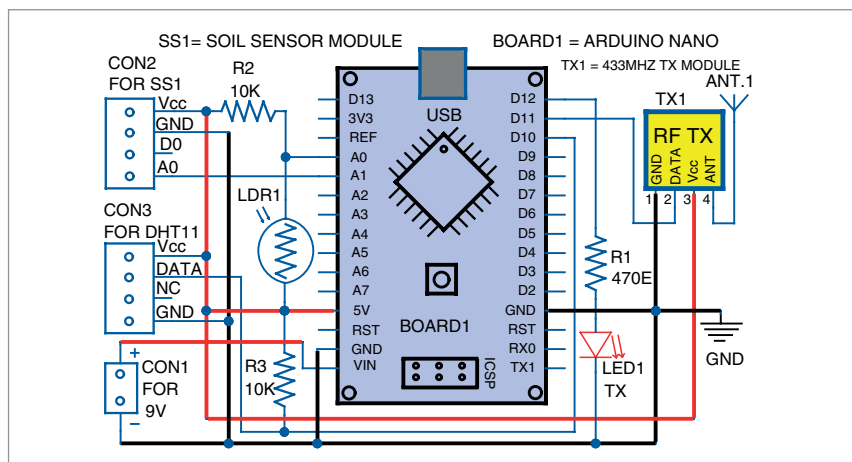


Fig. 2: Circuit diagram of a wireless sensor node

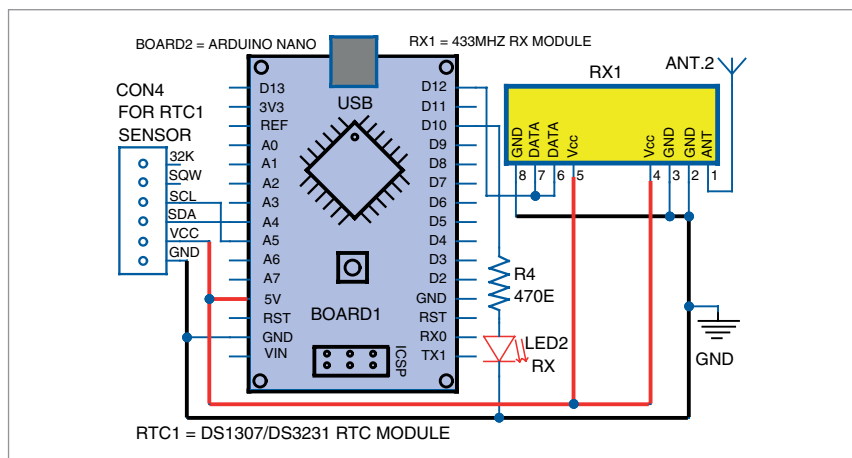


Fig. 3: Circuit diagram of remote data receiver

nected to digital pin D11 of Arduino board. An antenna (ANT.1) is formed by winding a 1mm diameter, 35cm long copper wire (single core) that is connected to antenna pin.

5. LED (LED1), which is connected to digital pin D12 through current limiting resistor R1.

A 9V battery or regulated power supply is used to power Arduino board. Arduino board Vin pin is connected to battery, so the board generates 5V supply from onboard 7805 voltage regulator chip, which is given to DHT11 and 433MHz RF transmitter module (TX1).

The sensor node periodically senses temperature, humidity, soil moisture, and ambient light and transmits the values. But, as shown in the block diagram, there may be any number of sensor nodes. They would all be transmitting their values. It may happen that more than one node

transmit data at the same time. Also, how the receiver would know the data received is from which node?

To overcome these problems all sensor nodes are synchronised in time such that two nodes do not transmit simultaneously. But if that happens, the receiver will accept any one sensor data. Also, all the nodes are assigned numbers (say, from 1 to N). Before they transmit their data, they transmit their node number. So, the receiver knows the exact sensor node from which the data is coming.

The DHT11 sensor gives accurate and calibrated measurements of temperature and humidity. It gives direct digital values for both measurements. Arduino microcontroller reads these values and converts them into character form (ASCII).

As ambient light falling on LDR1 increases, its resistances decreases and analogue voltage output increases.